

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Main)-2024

TEST DATE: 11-01-2024

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

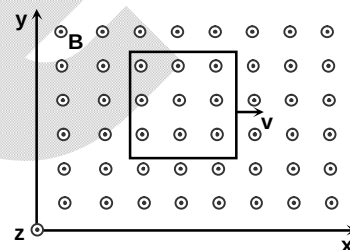
PART – A

SECTION – A (One Options Correct Type)

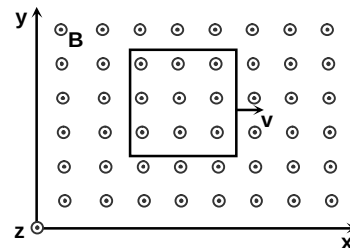
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A black box is connected across an ac source of emf $V = 100\sin(100s^{-1}t)$ in volt. Phase of the current in the circuit is ahead of the phase of the source voltage by $\frac{\pi}{4}$. The peak current is observed to be $\sqrt{2}A$. The black box consists of,
 - (A) Only a resistance of 50Ω
 - (B) Only a capacitance of $0.2mF$
 - (C) A series combination of a resistance of 50Ω and an inductance of $0.2mH$
 - (D) A series combination of a resistance of 50Ω and a capacitance of $0.2mF$

2. In a certain region of space magnetic field is present along positive z-axis. The magnetic field decreases with time at the rate of 1 T/s and also decreases along x-axis at the rate of 1 T/meter . A square conducting loop of side 10 cm and having electrical resistance 20Ω is lying in this region with two opposite sides parallel to x-axis and other two sides parallel to the y-axis. The loop is moving parallel to the x-axis with the constant speed of 2 m/s keeping its orientation same. Find the current induced in the loop.
 - (A) 2.5 mA
 - (B) 3.5 mA
 - (C) 5.2 mA
 - (D) 5.3 mA



3. In a certain region of space magnetic field is present along positive z-axis. The magnetic field decreases with time at the rate of 1 T/s and also decreases along x-axis at the rate of 1 T/meter . A square conducting loop of side 10 cm and having electrical resistance 20Ω is lying in this region with two opposite sides parallel to x-axis and other two sides parallel to the y-axis. The loop is move parallel to the x-axis with the constant speed of 2 m/s keeping its orientation same. Find the force needed to move it.



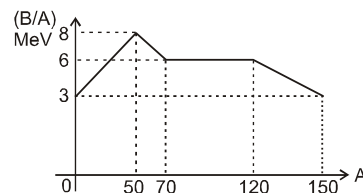
Find the force needed to move the loop with the constant speed given in the question.

- (A) $F = 1.5 \times 10^{-5}\text{ N}$
 - (B) $F = 0.5 \times 10^{-7}\text{ N}$
 - (C) $F = 1.5 \times 10^{-7}\text{ N}$
 - (D) $F = 2.0 \times 10^{-7}\text{ N}$
4. An EM wave is propagating in the positive x direction. The magnetic field is oscillating along the y-axis. At the origin magnetic field is given by the equation $B = 10^{-5}\text{ T} \sin(2\pi \times 10^{10}\text{ s}^{-1}t)$, where t is the time in second. Write the equation for the time varying electric field at distance x from the origin.
 - (A) $A = 10^3 \sin\left(2\pi \times 10^{10} - \frac{200}{3}\pi x\right)$
 - (B) $A = 2 \times 10^3 \sin\left(2\pi \times 10^{10} - \frac{200}{3}\pi x\right)$
 - (C) $A = 3 \times 10^3 \sin\left(2\pi \times 10^{10}t - \frac{200}{3}\pi x\right)$
 - (D) $A = 3 \times 10^3 \sin\left(2\pi \times 10^{10}t - \frac{100}{3}\pi x\right)$

5. Reflecting surface of a mirror is parallel to the x-y plane. An incident ray parallel to a vector $\hat{A} = \frac{1}{2\sqrt{2}}\hat{i} + \frac{1}{2\sqrt{2}}\hat{j} - \frac{\sqrt{3}}{2}\hat{k}$, falls on the mirror. Identify the direction of reflected ray. Express your answer in terms of $\hat{i}$, $\hat{j}$ and $\hat{k}$, where $\hat{i}$, $\hat{j}$ and $\hat{k}$ have their usual meanings.
- (A) $\frac{1}{2\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{\sqrt{3}}{2}\hat{k}$ (B) $\frac{1}{2\sqrt{2}}\hat{i} + \frac{1}{2\sqrt{2}}\hat{j} + \frac{\sqrt{3}}{2}\hat{k}$
 (C) $\frac{1}{2\sqrt{2}}\hat{i} + \frac{1}{2\sqrt{2}}\hat{j} + \sqrt{3}\hat{k}$ (D) $\frac{1}{2\sqrt{2}}\hat{i} + \frac{1}{2\sqrt{2}}\hat{j} + \frac{\sqrt{3}}{2\sqrt{2}}\hat{k}$
6. There exists a uniform and constant magnetic field $B = 1.67\text{T}$ along positive z-axis. Simultaneously a proton and an alpha particle are projected from the origin with same speed $v = 4\sqrt{5} \times 10^5 \text{ms}^{-1}$ along positive and negative x-axis respectively. Find the separation between them when their velocities become mutually perpendicular for the first time. Ignore the force of interaction between them as well as any force other than magnetic force. Mass of the proton is $1.67 \times 10^{-27} \text{kg}$.
- (A) 50 mm (B) 25 mm
 (C) 20 mm (D) 40 mm
7. At a given instant there are 25% undecayed radio- active nuclei in a sample. After 10 seconds the number of undecayed nuclei reduces to 12.5%. Mean-life of the nuclei is :
- (A) 14.43 sec (B) 13.73 sec
 (C) 10 sec (D) 20 sec
8. At a given instant there are 25% undecayed radio- active nuclei in a sample. After 10 seconds the number of undecayed nuclei reduces to 12.5%. The time in which the number of undecayed nuclei will further reduce to 6.25% of the reduced number is
- (A) 20 sec (B) 30 sec
 (C) 40 sec (D) 50 sec
9. Two identical large sheets of insulating material are uniformly charged with surface charge density $+\sigma$ each. Now the sheets are kept parallel to each other at a small separation with their planes parallel to the y-z plane of a coordinate system. The x coordinate of one of the plates is zero and that of the other is d. Read the following statements.
- (i) Potential at any point in the space between the sheets is equal to the potential at $x = 2d$.
 (ii) Potential at a point P on the positive x-axis at a distance x from the origin is smaller than the potential at origin by $\frac{\sigma}{\epsilon_0}(x - d)$
 (iii) In the space between the sheets the potential gradient is non-zero.
 (iv) All along the path on the positive x-axis the potential gradient is negative for $x > d$
- (A) only (i) is correct
 (B) only (ii) is correct
 (C) only (i), (ii) and (iii) are correct
 (D) (ii) and (iv) are correct
10. A polaroid is kept stationary with its plane parallel to the x-y plane. A plane polarized light polarized along the y-axis falls on the plane of the polaroid normally. The polaroid is rotated in its own plane. For the angular rotation 20° and 80° in same sense the intensity of the transmitted light is observed to be $\frac{3}{4}$ th of the maximum intensity. Find the initial inclination of the transmission axis of the polaroid with the y-axis may be
- (A) 30° (B) 90°
 (C) 130° (D) 150°

11. A disc of radius 2 m is lying at rest on a horizontal surface with plane horizontal. A small coin is placed on the disc at a distance of 1m from its centre. Now the disc starts rotating with constant angular acceleration of 1 rad/s^2 about a fixed vertical axis passing through the centre of the disc. Calculate the time when the coin starts slipping on the disc. The coefficient of friction between the disc and the coin is $\mu = \sqrt{0.17}$. Take $g = 10 \text{ ms}^{-2}$.
- (A) 1 s (B) 2 s
(C) 3 s (D) 1.5 s
12. A uniform string of linear mass density μ and length ℓ is rotated on a horizontal surface about a fixed vertical axis passing through one end of the string. The angular speed of the string is ω and its length remains perpendicular to the axis of rotation. The speed of a transverse pulse (with respect to the string) at the mid-point of the string is
- (A) $\frac{\omega \ell}{2}$ (B) $\frac{\omega \ell}{\sqrt{2}}$
(C) $\frac{\sqrt{3}\omega \ell}{2\sqrt{2}}$ (D) $\frac{\omega \ell}{8}$
13. A proton and an α -particle are accelerated by the same potential difference. The ratio of the d-Broglie wavelength of the proton to that of the α -particle is,
- (A) 1 (B) 1/4
(C) $2\sqrt{2}$ (D) 1/8
14. Under the influence of a periodic force F_1 a particle performs SHM along x-axis and its x-coordinate varies with time as, $x = A \sin \omega t$. When the force F_1 is replaced by another periodic force F_2 , particle performs SHM along y-axis and its y-coordinate varies with time as, $y = B \sin(\omega t + \Phi)$. When both forces are applied on the particle simultaneously, the particle performs SHM along a straight line in the x-y plane.
- (i) The value of Φ may be zero (ii) The value of Φ may be π
(iii) The slope of the resulting path may be $\frac{B}{A}$ (iv) The slope of the resulting path may be $-\frac{B}{A}$
- Choose the correct option.
- (A) Only (i) is correct (B) Only (ii) is correct
(C) Only (iii) is correct (D) All (i), (ii), (iii) and (iv) are correct.
15. A and B are two bulbs with rated powers are 200V, 100W and 250V, 125W respectively. The bulbs are connected in parallel across the same dc source. The ratio of the power consumed by the bulb A to the bulb B is,
- (A) 4/5 (B) 5/4
(C) 2/5 (D) 5/2
16. In a young's double slit experiment light of wavelength 600nm is used. P is the nearest point from the central maximum where intensity is $3/4^{\text{th}}$ of the maximum intensity. What percentage of the maximum intensity is observed at point P if light is replaced by the wavelength 400 nm?
- (A) 50% (B) 75%
(C) 25% (D) 100%

17. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Consider a nucleus of $A = 110$. Fission of this nucleus results into 2 fragments. Possible mass number of the resulting nuclei may be
- (A) 55 and 55
(B) 70 and 40
(C) 90 and 20
(D) 80 and 30



18. Two simple pendulum of length ℓ and 16ℓ are released from the same phase together. They will be at the same phase after a minimum time.

(A) $\frac{8\pi}{3} \sqrt{\frac{\ell}{g}}$

(B) $\frac{\pi}{3} \sqrt{\frac{\ell}{g}}$

(C) 2 s

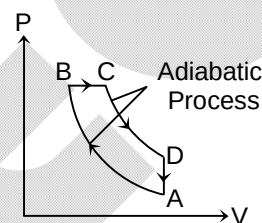
(D) none of these

19. A cyclic process perform by a heat engine is shown in the figure. Figure shows that

- (i) Heat is absorbed in the process BC.
(ii) Heat is released in the process DA.

Choose the correct option(s).

- (A) Only (i) is correct
(B) Only (ii) is correct
(C) Both (i) and (ii) are correct
(D) Both (i) and (ii) are wrong



20. In a conservative field motion of a particle of mass 2 kg is confined along x-axis. The potential energy of the particle varies with its x coordinate according to the function, $U(x) = ax^4 - bx^2$, where, x is the distance in meter from the origin, $a = 1 \text{ J m}^{-4}$ and $b = 8 \text{ J m}^{-2}$. Initially, particle is at rest at the origin, and due to slight disturbance, it starts moving along positive x-axis. Calculate the maximum speed acquired by the particle during motion is
- (A) 2 m/s
(B) 4 m/s
(C) 8 m/s
(D) 1 m/s

SECTION – B

(Numerical Answer Type)

This section contains 10 Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A rigid wire is stationary and inclined at an angle $\theta = 30^\circ$ with the horizontal. A bead of mass $M = 9 \text{ kg}$ is free to slide along the length of the wire is held stationary on the wire at some height. A simple pendulum having small bob of mass $m = 4 \text{ kg}$ is suspended from the bead. Initially system is stationary, and string of the pendulum is vertical. The accelerations is $\frac{N}{2} \text{ m/s}^2$ of the bead as soon as system is set free. Then value of N is
22. In a helium ion an electron is orbiting the nucleus in the nth orbit. An EM wave of wave length $900\text{\AA}$ ionises the atom. Find the value of n. Take $hc = 1224 \text{ eVnm}$
23. Two spheres A and B having radii 3 cm and 5 cm respectively are coated with carbon black on their outer surfaces. The wavelengths of maximum Intensity of emission of radiation are 300 nm and 500 nm respectively. The respective powers radiated by them are in the ratio of $\frac{25}{K}$. Find the value of K.

24. To remove an object from earth's gravitational field the speed that is to be given to it on the surface of the earth is nearly equal to 11.25 km/s. However, when an object on the earth is to be removed from the solar system, gravitational influence of the sun is taken in account. To remove an object from the solar system, the speed that is given to it on earth's surface is nearly equal to 45 km/s. The mass of the sun is k_1 times the mass of the earth and the orbital radius of the earth is k_2 times its radius. Find the value of k_1/k_2 .
25. A total charge Q is uniformly distributed throughout the volume of a sphere. Electric potential at a point on the surface of the sphere is greater than the potential at infinity by 16 V. Find the potential difference (in volt) between point **P** at a distance half of the radius of the sphere from the centre and the point **S** at a distance double the radius of the sphere from the centre of the sphere.
26. A block of mass 1 kg is released from rest on an inclined plane of inclination 30° . The block acquires the kinetic energy of 5J when slides down the plane by 2 m. How much force (in newton) is needed to be applied on the block to move up the plane so that it will gain the same kinetic energy over same path length? Take $g = 10 \text{ m/s}^2$.
27. A body cools down in the open air as per Newton's law of cooling. Temperature of the body falls from 80°C to 70°C in 9 minute. How long time (in minute) further will it take the body to fall its temperature from 70°C to 60°C . The temperature of the surrounding air is constant at 20°C .
28. A sonometer wire resonates with frequency f_0 in its fundamental mode. If length of the wire is decreased to the value 0.95 m, the resonating frequency increases by 5 Hz and if length is increased to the value 1.0 m, the resonating frequency decreases by 5 Hz. Find the length (in cm) of the wire for the resonating frequency f_0 . The speed of the wave in the wire is 382.2 m/s.
29. A parallel capacitor of plate area $A = 4 \times 10^{-4} \text{ m}^2$ and separation between the plates $d = 1.1 \text{ mm}$ is connected across a cell of emf $V = 10 \text{ V}$. the space between the plates is completely filled with an insulating material of dielectric of constant $K = 100$. The polarisation of the dielectric material is, $k \times 10^{-19} \text{ C/m}^2$. Find the value of $5k$.
30. A lens with one surface concave and the other surface convex is made up of glass of refractive index 1.5. Radius of curvature of the concave surface is 50 cm and that of the convex surface is 25 cm. The convex surface is silvered. The silvered lens and an object are kept inside water of refractive index $4/3$. The object is placed at $\frac{N}{17} \text{ cm}$ in front of the concave surface so that its image will coincide with the object. Find the value of N .

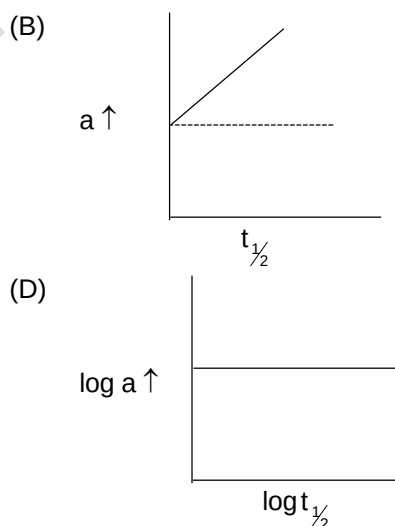
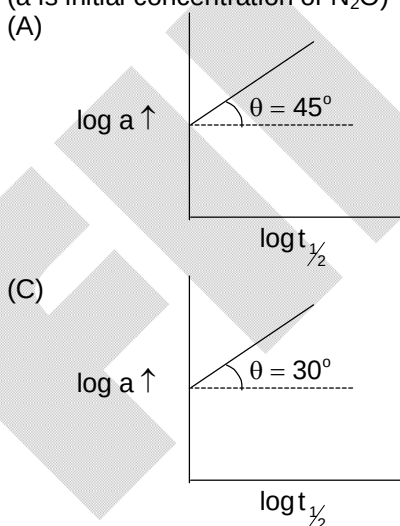
Chemistry

PART – B

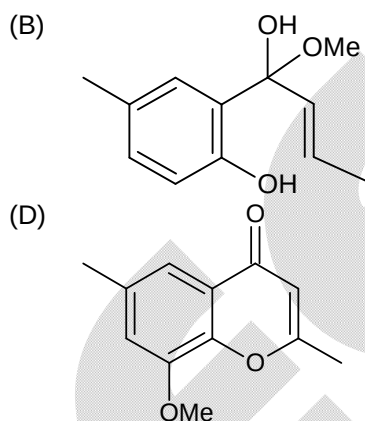
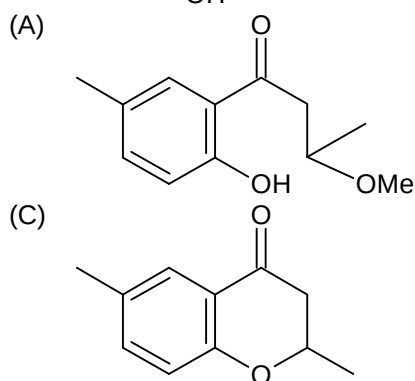
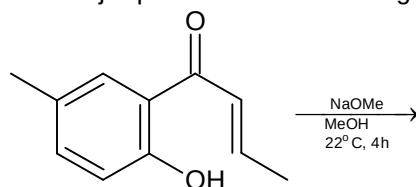
SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

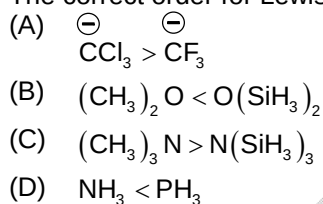
31. Which of the following is not water soluble vitamin?
 (A) Thiamine (B) Riboflavin
 (C) Vitamin C (D) Vitamin A
32. Dimethylformamide $\xrightarrow[(2) H^+]{(1) PhLi}$ X (degree of unsaturation=5)
 Which of the following statement about X is **correct**?
 (A) X on oxidation gives ketone.
 (B) X reacts with iodoform reagent to produce an intense yellow precipitate.
 (C) X reacts with Tollen's reagent, producing a silver mirror as an observable product.
 (D) X on reaction with ceric ammonium nitrate forms a red colour precipitate.
33. The first ionization energy for which of the following elements is highest?
 (A) Tl (B) Ga
 (C) Al (D) In
34. Which of the following is not obtained when $KMnO_4$ is heated at 513 K?
 (A) K_2MnO_4 (B) MnO
 (C) MnO_2 (D) O_2
35. Which of the following is correct graph for the reaction?
 $2N_2O \xrightarrow{Pt(hot)} 2N_2 + O_2$
 (a is initial concentration of N_2O)



36. The major product of the following reaction is



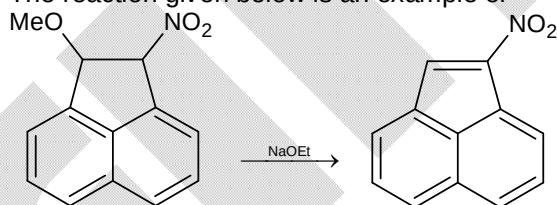
37. The correct order for Lewis base strength is



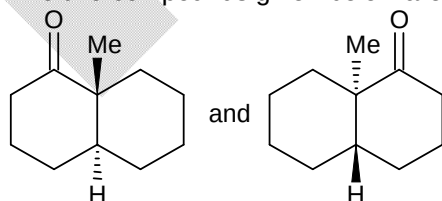
38. Which is the correct order of bond angle?

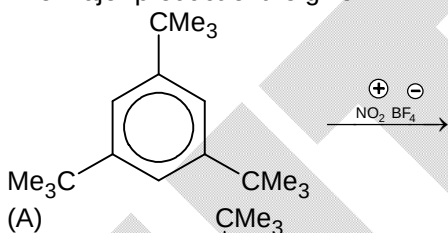


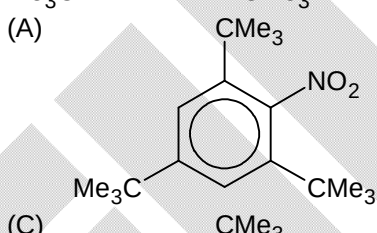
39. The reaction given below is an example of

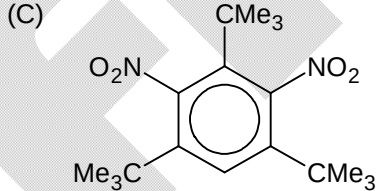


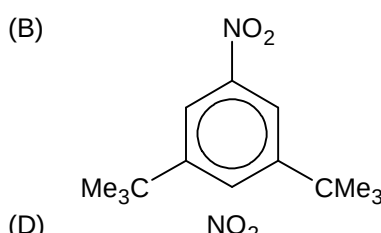
40. The two compounds given below are

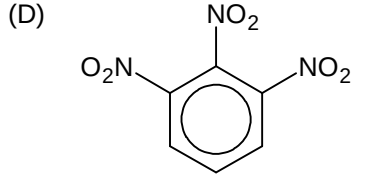


41. The red colour of ruby is due to
 (A) d – d transition of Cr^{3+} ion in Cr_2O_3 lattice
 (B) d – d transition of Cr^{3+} ion in Al_2O_3 lattice
 (C) ligand to metal charge transfer transition
 (D) metal to ligand charge transfer transition
42. The complex that absorbs light of shortest wavelength is
 (A) $[\text{CoF}_6]^{3-}$
 (B) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 (C) $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (D) $[\text{Co}(\text{ox})_3]^{3-}$
43. Kohlrausch's law is applicable to a infinite dilute solution of
 (A) KCl in hexane
 (B) acetic acid in benzene
 (C) HCl in water
 (D) benzoic acid in benzene
44. Upon heating a 500 ml of semimolar HCl solution, 1.335 g of HCl is lost. The molarity of resultant solution is 0.569 M. The volume of the solution shrinks to _____ ml.
 (A) 375 ml
 (B) 750 ml
 (C) 500 ml
 (D) 700 ml
45. The sodium fusion extract is treated with sodium nitroprusside, appearance of a coloured complex indicates presence of sulphur. What is the colour and formula of complex?
 (A) Red, $[\text{Fe}(\text{SCN})]^{2+}$
 (B) Blue, $\text{Fe}_4 [\text{Fe}(\text{CN})_6]_3$
 (C) Violet, $[\text{Fe}(\text{CN})_5 \text{NOS}]^{4-}$
 (D) Blue, $[\text{Fe}(\text{CN})_5 \text{NO}]^{2-}$
46. The major product of the given
- 

(A) 

(C) 

(B) 

(D) 
47. Which of the following statement is **incorrect**?
- (A) The melting point of hydrides of Group 15 : $\text{NH}_3 > \text{SbH}_3 > \text{AsH}_3 > \text{PH}_3$
 (B) ΔH_{eg} kJ mol^{-1} : $\text{S} > \text{Se} > \text{O} > \text{Te} > \text{Po}$
 (C) Melting point : $\text{B} > \text{Al} > \text{Tl} > \text{In} > \text{Ga}$
 (D) Bond enthalpy : $\text{C} - \text{C} > \text{Si} - \text{Si} > \text{Ge} - \text{Ge} > \text{Sn} - \text{Sn}$

48. When 2g non-volatile hydrocarbon containing 99.4% carbon by mass is dissolved in 100g of benzene, the vapour pressure of benzene at 30°C is lowered from 89.78 mm Hg to 89.0 mm Hg. The molecular formula of the hydrocarbon is
 (A) $C_{12}H_{34}$ (B) $C_{13}H_{22}$
 (C) $C_{14}H_{12}$ (D) $C_{14}H_{10}$
49. The oxidation number of sodium in sodium amalgam is
 (A) +2 (B) +1
 (C) -2 (D) 0
50. The ion that cannot be precipitated by using HCl and H_2S is
 (A) Pb^{2+} (B) Ni^{2+}
 (C) Ag^+ (D) Sn^{2+}

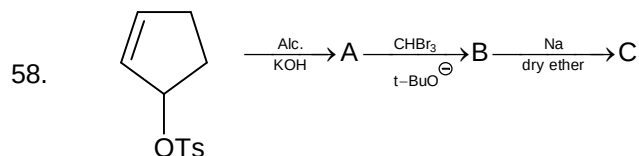
SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

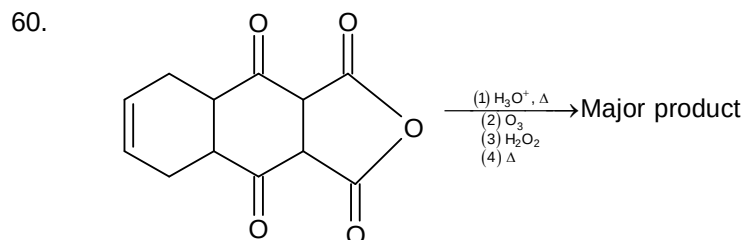
51. The equation of state for one mole of a gas is $PV = RT + BP$, where B is a constant, independent of temperature. The internal energy is the function of temperature only. The enthalpy change of a gas is approximately - 332 J, if one mole of above gas is isothermally expanded from x L to 22 L at a constant external pressure of 1 bar at 400 K. The x is apparently equal to _____ (B = 2 L/mol).
52. At 200°C, PCl_5 dissociates as $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. The percentage dissociation of PCl_5 at 200°C is 68.1%. The equilibrium vapour are _____ times as heavy as hydrogen.
53. A particular water sample has x ppm $CaSO_4$. 75% of water by mass must be evaporated in a container before solid $CaSO_4$ begins to deposit. Assume that the solubility of $CaSO_4$ does not change with temperature in the range 0°C to 100°C. The value of K_{sp} for $CaSO_4$ is 1.6×10^{-5} . The value of x is _____ (Consider density of water sample is 1 g/mL)
54. The H_2 is dissociated by illumination with radiation of wavelength 270 nm then 2.5% percentage of radiant energy will be converted in to kinetic energy. What is the dissociation energy of H_2 per mole in kJ? ($h = 6 \times 10^{-34}$ Js and $N_A = 6 \times 10^{23}$)
55. The sum of geometrical isomers in $[Pt(H_2N-CH(CH_3)COO^-)_2]$ complex and stereoisomers of $[Pt(gly)_3]^+$ complex is _____.
56. How many of the transition elements are called platinum group metals?
57.
$$Ph-CH=CH-\underset{\substack{| \\ CH_3}}{CH}-CHO \xrightarrow[pH=4.5]{NH_2OH} \text{Product}(x)$$

In the above reaction product formed is x. The ratio of the sum of possible number of stereoisomers and stereocentres of the product to the number of chiral centres in the product x is _____



The number of degree of unsaturation in A + B + C is

59. The sum of bond order and number of π – bonds in C_2 molecule on the basis of molecular orbital theory is _____.



The ratio of number of carbons in the product to the number of keto groups in the product is

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let $A = \{1, 2, 3, \dots, 2024\}$. Number of subsets of the set A such that sum of all the elements is even is
 (A) 2^{2023} (B) $2^{2023} - 1$
 (C) 2^{2022} (D) none of these
62. Let $R = \{(x, y) : (x, y) \text{ is equidistant from the lines } y - \sqrt{3}x = 0 \text{ and } \sqrt{3}y - x = 0\}$, then R is
 (A) reflexive and symmetric but not transitive (B) symmetric and transitive but not reflexive
 (C) reflexive and transitive but not symmetric (D) equivalence relation
63. For $x \in (\sqrt{3}, 3)$ let $f(x) = \tan x$, $g(x) = x^{1/4}$, $h(x) = \frac{x^4 + \sqrt{3}}{1 - \sqrt{3}x^4}$ and $(\text{hogof})(x) = p(x)$, then $p\left(\frac{5\pi}{7}\right)$ equals
 (A) $\tan\left(\frac{22\pi}{21}\right)$ (B) $\tan\left(\frac{8\pi}{21}\right)$
 (C) $\tan\left(-\frac{22\pi}{21}\right)$ (D) $\tan\left(-\frac{8\pi}{21}\right)$
64. $\lim_{x \rightarrow 0^+} \left[\frac{a}{x} \right] \left[\frac{b}{x} \right] \left[\frac{x}{c} \right] \left[\frac{x}{d} \right]$ (a, b, c, d > 0) (where [.] represents greatest integer function)
 (A) equals ab (B) equals 0
 (C) equals $\frac{ab}{cd}$ (D) does not exist
65. Let $f(x) = [\sin(\pi x)] \{\cos(\pi x)\}$; $x \in (0, 100)$. Number of points where f(x) is discontinuous is (where [.] represents greatest integer function and {.] represents fractional part function)
 (A) 98 (B) 99
 (C) 100 (D) none of these
66. Let $ax^2 + bx + c = 0$ and $a'x^2 + b'x + c' = 0$ be two quadratic equations having roots α, β and γ, δ respectively. If all the coefficients are real and $\frac{\alpha^{2024} + \beta^{2024}}{\alpha^{-2024} + \beta^{-2024}} + \frac{\gamma^{2024} + \delta^{2024}}{\gamma^{-2024} + \delta^{-2024}} = 0$, then which of the following must be true?
 (A) $a + a' = 0$ (B) $b + b' = 0$
 (C) $c + c' = 0$ (D) none of these
67. Largest number of the following numbers which divides $(2025^{2024} - 2021^{2024})$ is
 (A) 4 (B) 8
 (C) 16 (D) 32

68. $\lim_{x \rightarrow \infty} \frac{x^5 \left[(\sqrt{2x+3} + \sqrt{2x-5})^{10} + (\sqrt{2x+3} - \sqrt{2x-5})^{10} \right]}{(2x + \sqrt{x^2 - 3})^{10} + (2x - \sqrt{x^2 - 3})^{10}}$ equals
- (A) $\left(\frac{2}{3}\right)^{10}$ (B) $\left(\frac{2}{3}\right)^5$
 (C) $\frac{2^{15}}{3^{10}}$ (D) none of these
69. Let $f : \left(0, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ and $g : \left(0, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be two functions defined by $f(x) = \frac{\sin x}{x}$ and $g(x) = \frac{\tan x}{x}$, then $f(g(1-\alpha)) > f(g(\alpha^2 - 1))$ holds for which value of α ?
- (A) $-\frac{21}{20}$ (B) $\sqrt{\frac{\pi}{2}}$
 (C) $-\sqrt{\frac{\pi}{2}} + 1$ (D) none of these
70. Let $f(x) = |e^{2x} - e^x| + [e^{2x} - e^x]$; $x < 0$. If $f(x) \in (a, b]$ then $a + b$ is equal to (where $[.]$ represents greatest integer function)
- (A) $-\frac{3}{4}$ (B) $-\frac{7}{4}$
 (C) -1 (D) none of these
71. $\int_0^{\pi/2} x \sin x \, dx$ equals
- (A) $\frac{\pi^2}{8} - \frac{\pi^4}{384} + \int_0^{\pi/2} \frac{x^5 \sin x}{120} \, dx$ (B) $\frac{\pi^2}{8} - \frac{\pi^4}{384} + \int_0^{\pi/2} \frac{x^5 \sin x}{20} \, dx$
 (C) $\frac{\pi^2}{8} - \frac{\pi^4}{64} + \int_0^{\pi/2} \frac{x^5 \sin x}{20} \, dx$ (D) $\frac{\pi^2}{8} - \frac{\pi^4}{64} + \int_0^{\pi/2} \frac{x^5 \sin x}{120} \, dx$
72. Let $f(x) = \tan x$, $x \in \left(0, \frac{\pi}{2}\right)$ and $g(x) = \operatorname{cosec} x$, $x \in \left(0, \frac{\pi}{2}\right)$. Area bounded by $y = f(x)$, $y = g(x)$ and $x = \frac{\pi}{6}$ is
- (A) $\ln\left(\frac{18-8\sqrt{5}}{\sqrt{5}-1}\right) - \ln\left(\frac{2\sqrt{3}-3}{2}\right)$ (B) $\frac{1}{2} \ln\left(\frac{18-8\sqrt{5}}{\sqrt{5}-1}\right) - \ln\left(\frac{2\sqrt{3}-3}{2}\right)$
 (C) $\ln\left(\frac{18-8\sqrt{5}}{\sqrt{5}-1}\right) - \frac{1}{2} \ln\left(\frac{2\sqrt{3}-3}{2}\right)$ (D) none of these
73. Let AB, BC and AC be tangents to a circle S where equation of AB is $x + y = 3$. If S touches AB at (1, 2) and AC at (3, 5), then equation of AC is
- (A) $7x + 17y - 106 = 0$ (B) $13x - 8y + 1 = 0$
 (C) $5x - 3y = 0$ (D) none of these

74. If the equation of directrix of a rectangular hyperbola is $x + y = 1$; whose centre is $(0, 5)$; is $ax^2 + by^2 + 4xy + cx + dy + e = 0$ then $c + d + 2e$ equals
 (A) -58 (B) $-46 + 4\sqrt{2}$
 (C) $-46 - 4\sqrt{2}$ (D) -84
75. The equation of two sides of a variable triangle are coordinate axes. If the third side is tangent to $y = \frac{1}{x}$, then locus of circumcentre of the triangle is P. Which of the following points lie on P?
 (A) $(1, 1)$ (B) $(2, 2)$
 (C) $(-1, 2)$ (D) $(2, -1)$
76. Let AB be the shortest distance between the lines
 $L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-2}{2}$ and $L_2 : \frac{x-2}{1} = \frac{y-2}{2} = \frac{z-3}{3}$
 where A lies on L_1 and B on L_2 . Coordinates of B are
 (A) $(\frac{5}{2}, 3, \frac{9}{2})$ (B) $(\frac{29}{13}, \frac{32}{13}, \frac{45}{13})$
 (C) $(\frac{32}{13}, \frac{38}{13}, \frac{57}{13})$ (D) none of these
77. Let A, B, C be three events. Given below are two statements
 $S_1 : P\left(\frac{A}{C}\right) + P\left(\frac{B}{C}\right)$ can be greater than 1
 $S_2 : P\left(\frac{A \cup B}{C}\right)$ can be greater than 1,
 then
 (A) only S_1 is true (B) only S_2 is true
 (C) both S_1 and S_2 are true (D) neither S_1 nor S_2 is true
78. Let ABC be a triangle and DEF be points on the sides of the triangle such that
 $\frac{BD}{DC} = \frac{1}{2}$, $\frac{CF}{FA} = \frac{1}{3}$ and $\frac{AE}{EB} = \frac{1}{4}$, then $\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle DEF)}$ equals
 (A) $\frac{25}{12}$ (B) $\frac{12}{5}$
 (C) $\frac{11}{6}$ (D) none of these
79. Let there be 9 observations such that every observation is repeated atleast once and the set of observations form a G.P. with common ratio 2. The median and mode of 9 observations are 8 and 4 respectively. If 4 smallest observations are increased by 10 each then new mode is
 (A) 14 (B) 18
 (C) 26 (D) cannot be determined with given data
80. Number of solutions of the equation $\sin^{-1}(x^4 - 8x^2 + 17) + \cos^{-1}(x^2 - 3x + 2) = \sec^{-1}(x - 3)$ is
 (A) 0 (B) 1
 (C) 4 (D) 16

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let $S = \{z \in \mathbb{C} : z + i\bar{z} = |z|^2\}$ then maximum value of $4|z|^2$ equals to
82. Let S be the set of all 3×3 matrices that can be formed using either 0 or 1. The trace of matrix $\sum_{A \in S} A$ equals
83. Number of values of 'P' such that the system of equations

$$\begin{aligned} Px + y + z &= 0 \\ x + Py + z &= 1 \\ x + y + z &= 2 \end{aligned}$$
is inconsistent is
84. The number of 3 digit numbers that can be formed such that the absolute difference of adjacent digits is one unit is
85. Let $x, y, z > 0$ if $x^5 y z^3 = (75)^7$ then least value of $3x + y + z$ is
86. If $y = (x^x)^{x^x}$ and $\frac{dy}{dx} = \frac{y \ln y}{\ln x} \left[(\ln x)^2 + p \ln x + \frac{1}{x} \right]$ where $p = \frac{ax+b}{cx}$ (a, b, c are constants) then $a + b + c$ equal
87. A balloon is being filled with air at a rate of $50 \text{ cm}^3/\text{min}$. Assuming it always maintains a spherical shape, the diameter of the balloon when rate of change of volume equals rate of change of its surface area is (in cms)
88. Let a curve $y = y(x)$ be given by the solution of the differential equation
 $\sin(\tan^{-1}(e^x)) dx = (e^x + e^{-x}) dy$. If it passes through origin and $y(\ln \sqrt{3}) = \frac{\sqrt{2} - \beta}{\gamma}$, then $\beta + \gamma$ equals ($\beta, \gamma \in \mathbb{I}$)
89. If $\int \frac{x^4 + x^3 - x - 1}{x^3} e^{\left(x + \frac{1}{x}\right)} dx = \frac{g(x)}{x} e^{\left(x + \frac{1}{x}\right)} + c$, where $g(x)$ is a polynomial in x then $g(1)$ equals
90. Let $A(-1, 1)$ and $B(1, 3)$. C is a variable point which lies at a distance of 4 units from origin. If Δ_1 and Δ_2 be the minimum and maximum possible area bounded by the line segments AB, BC and AC respectively, then $\Delta_2 - \Delta_1 - 4\sqrt{2}$ equals